Project Workflow: Week 2

Phase: Requirements Analysis & System Design

1. Objective

The primary goal of Week 2 was to define the technical requirements and organizational structure of the Treasure Hunt game, ensuring that all gameplay elements and system logic were clearly established before starting development.

2. Detailed Task Breakdown

- **Requirement Finalization:** Analyzed and documented the core gameplay rules, including procedural generation logic, item interaction mechanics, and scoring criteria.
- **System Organization:** Established the hierarchy of game modules, ensuring that the exploration logic, player controls, and asset management systems are clearly separated for efficient development.
- **Game State Planning:** Outlined the operational logic for the various stages of the game, such as initialization, active exploration, and completion states, to ensure smooth transitions between events.

3. Deliverables

- **Functional Requirements Summary:** A detailed list of game mechanics, rules, and player objectives.
- **Modular Architecture Plan:** A structured overview of how the game's core files and components will function together.
- **Game Logic Overview:** A descriptive sequence of operations for handling player input and environmental updates.

4. Tools & Resources

- **Development Strategy:** Adoption of modular design principles to maintain a clean and scalable codebase.
- **Documentation:** Project journal used for tracking requirement updates and logical progression.

Core System Modules

Module Name	Primary Responsibility
Game Controller	Manages the primary game loop, timing, and overall application state.
Map Engine	Handles the procedural generation of the island environment and terrain layout.
Player Module	Manages character movement, torch fuel consumption, and inventory state.
Asset Handler	Manages the loading and management of game resources and visual elements.
Scoring System	Monitors player actions to calculate and update scores based on performance.